Factorial :

↳ n! is continuous multiplication of first n-natural numbers.

$4! = 1\times2\times3\times4$

$6! = 1\times2\times3\times4\times5\times6$

ⓐ $0! = 1! = 1$

Fundamental principle of Counting :

(I) Multiplication Principle (AND) :→

Jab tak process complete na ho tab tak taste mein aane wale sabhi no. of ways ko multiply karte raho.

(II) Addition Principle (OR) :→

Agar given process 'm' tarike se complete ho gaya hai ~~and~~ and uss ke process ko complete karne ke 'n' aur dusre tarike hai, then

Total no. of ways = m+n

Eg : A & B are two cities connected by 3 routes & B & C are connected by two different routes.

(I) In how many ways can a person go from A to C via B?

Sol^n



No. of ways = 3×2=6

(II) In how many ways can a person exit city B?

No. of ways = 3+2 = 5

Ques Total 7 man are gathered in conference hall. How many handshakes are possible?

Sol^n Hand shake b/w two men will occur only once so don't double count.



No. of ways = 6+5+3+2+1

= 21 Ans

Formation of Numbers :

Divisibility Rules

1) By 2 ⇒ Last digit must be even (0,2,4,6,8)

2) By 3 ⇒ Sum of all digits must be divisible by 3

3) By 4 ⇒ Last two digits must be divisible by 4

4) By 5 ⇒ Last digit must be 0 or 5

5) By 6 ⇒ (Divisible by 2) ∩ (Divisible by 3)

6) By 11 ⇒ (Sum of digits at even places) − (Sum of digits at odd places) must be divisible by 11.

Permutation & Combination-1

7) By 8 ⇒ Last 3 digits divisible by 8

8) By 25 ⇒ Last 2 digits divisible by 25

Ques How many 3 digit numbers can be formed by the digits 1,2,3,4,5

(I) w/o repetition

Sol^n ___ ___ ___ ⇒ 5 × 4 × 3 ⇒ 60 Ans

(II) with repetition

Sol^n ___ ___ ___ ⇒ 5 × 5 × 5 ⇒ 125 Ans

Ques How many 6 digits odd no. greater than 6,00,000 can be formed from 5,6,7,8,9,0 if repetition is allowed?

Sol^n (6,7,8,9) ↓ ___ ___ ___ ___ ___ ↓ (5,7,9)

4 × 6 × 6 × 6 × 6 × 3

= 15552 Ans

Ques Same above question but without repetition.

Sol^n Case-01 ⇒ ___ ___ ___ ___ ___ 5

4 × 4 × 3 × 2 × 1

Case-02 ⇒ ___ ___ ___ ___ ___ 1

3 × 4 × 3 × 2 × 1

Case-03 ⇒ ___ ___ ___ ___ ___ 9

3 × 4 × 3 × 2 × 1

Total numbers = 96 + 72 + 72

= 240 Ans

Rank of a Word in Dictionary :

When no alphabet is repeated ⇒

Step-1 ⇒ Assign numbers to all alphabets acc. to alphabetical order.

Step-2 ⇒ Below every alphabet, write total no. of alphabets with smaller digit than their's.

Step-3 ⇒ Now, from left to right, write 0! to n! factorial till the start of word.

Step-4 ⇒ Multiply both the rows and add.

Step-5 ⇒ Lastly, add 1 to the addition.

Ques Find Rank of word 'MATHS'.

Sol^n

③	①	⑤	②	④
M	A	T	H	S
2	0	2	0	0
4!	3!	2!	1!	0!

4!(2) + 0 + 2(2!) + 0 + 0 + 1 = 53 Ans

Ques Find Rank of word MOTHER.

Sol^n

③	④	⑥	②	①	⑤
M	O	T	H	E	R
2	2	3	1	0	0
5!	4!	3!	2!	1!	0!

2(5!) + 2(4!) + 3(3!) + 2! + 1

= 240 + 48 + 18 + 2 + 1 = 309 Ans

When alphabets are repeated ⇒

Follow the same step. Just remember to divide row 1 by factorial of the no. of alphabets repeating (including the starting alphabet as well).

Ques Find rank of word INDIA.

Sol^n

③	④	②	⑤	①
I	N	D	I	A
2	3	1	1	0

Including the 'I' first, there are two repeating alphabets → 2!

4! 3! 2! 1! 0!

24 + 18 + 2 + 1 + 0 + 1 = 46 Ans

Repeating alphabets are assigned same no. while numbering alphabetically.

Ques Find rank of JIT-JEE.

Sol^n

②	②	④	③	①	①
I	I	T	J	E	E
2	2	3	2	0	0
2!	2!	2!	2!	2!	2!

5! 4! 3! 2! 1! 0!

60 + 24 + 9 + 2 + 0 + 0 + 1 = 96 Ans

Ques The letters of COCHIN are permuted and all permutations are arranged like a dictionary. No. of words appearing before COCHIN are ___

Sol^n

①	⑤	①	②	③	④
C	O	C	H	I	N
0	4	0	0	0	0
2!					

5! 4! 3! 2! 1! 0!

4 × 24 ⇒ 96 words are there before COCHIN

Ques The total no. of numbers, lying b/w 100 and 1000 that can be formed with the digits 1,2,3,4,5, if the repetition of digits is not allowed and numbers are divisible by either 3 or 5 is ___

Sol^n Case-01 ⇒ No. is divisible by 5 ⇒

___ ___ 5 = 12

4 × 3

Case-02 ⇒ No. is divisible by 3 ⇒

(1,2,3) ⇒ 3!

(3,4,5) → 3!

(1,3,5) ⇒ 3!

(2,3,4) ⇒ 3!

But here we also count numbers ending with 5 which are already counted in case-01, so we minus them

Total such numbers

= 12 + 24 − 2 − 2

= 32 Ans

Finding Powers of primes :

↳ Iss bhi prime ke power nikalni hai, Uss prime no. se given number ko successively divide karte raho until we get (0.___), then take Greatest integer [.] of all & add them.

Eg: 100!

Soln $100! = 2^x \cdot 3^y \cdot 5^z \cdot \dots$

$\frac{100}{2} = [50]$ $\quad \frac{12.50}{2} = [6.25]$ $\quad \frac{1.__}{2} = [0.__]$

$\frac{50}{2} = [25]$ $\quad \frac{6.25}{2} = [3.125]$

$\frac{25}{2} = [12.50]$ $\quad \frac{3.125}{2} = [1.__]$

Power of 2 = 50+25+12+6+3+1 = 97

$\frac{100}{5} = [20]$ $\quad \frac{4}{5} = [0.8]$ ⋮ Power of 5 = 20+4+0 = 24

$\frac{20}{5} = [4]$

Hence, there will be max. 24 zeroes ($2^{24} \cdot 5^{24}$) in 100!

Ques Find no. of zeroes in 500!

Soln $\frac{500}{2} = 250 \to 125 \to 62.25 \to 31.125$

$0.__ \leftarrow 1.__ \leftarrow 3.__ \leftarrow 7.__ \leftarrow 15.__$

Power of 2 = 250+125+62+31+15+7+3+1 = 494

$\frac{500}{5} = 100 \to 20 \to 4 \to 0.8$

Power of 5 = 100+20+4 = 124

No. of zeroes = 124 Ans

Permutation v/s Combination :

Arrangement ↙ ↘ Selection

$^nP_r = \frac{n!}{(n-r)!}$ $\qquad$ $^nC_r = \frac{n!}{r!(n-r)!}$

$^nP_r = {}^nC_r \times r!$

Ques If $^1P_1 + {}^2P_2 \cdot 2 + {}^3P_3 \cdot 3 \dots + {}^{15}P_{15} = {}^zP_y - s$, $0 \le s \le 1$, then find ~~$^{z+s}C_{y-s}$~~

Soln $\because {}^nP_n = \frac{n!}{0!} = n!$

$S = 1! + 2(2!) + 3(3!) + \dots 15(15!)$

$= \sum_{r=1}^{15} r(r!) \Rightarrow \sum_{r=1}^{15} (r+1-1)r!$

$S = \sum_{r=1}^{15} (r+1)r! - \sum_{r=1}^{15} r!$

$\Downarrow (r+1)!$

$T_1 = \cancel{2!} - 1!$

$T_2 = \cancel{3!} - 2!$

$T_3 = \cancel{4!} - 3!$

$\vdots$

$T_{15} = 16! - \cancel{15!}$

$S = 16! - 1!$

$= {}^{16}P_{16} - 1$

$\Rightarrow \boxed{q = p = 16}$ $\quad \boxed{s = 1}$

So, $^{q+s}C_{y-s} = {}^{16+1}C_{16-1} \Rightarrow {}^{17}C_{15}$

$= \frac{17 \times 16 \times 8}{1 \times 2} \Rightarrow 136$ Ans

Ques The value of $(2 \cdot {}^1P_0 - 3 \cdot {}^2P_1 + 4 {}^3P_1 - \dots$ upto 51 terms) + $(1! - 2! + 3! - \dots$ upto 51 terms) is equal to ____.

Soln $[2(1!) - 3(2!) + 4(3!) \dots + 52(51!)]$

$+$

$[1! - 2! + 3! - 4! \dots + 51!]$

$\Downarrow$

$[\cancel{2!} - \cancel{3!} + \cancel{4!} - \cancel{5!} \dots + 52!]$

$+$

$[1! - \cancel{2!} + \cancel{3!} - \cancel{4!} \dots + \cancel{51!}]$

$\Rightarrow \boxed{52! + 1}$ Ans

Theorem-01 :→

No. of combination of n-distinct things taken r at a time when p particular things are always included = $\boxed{{}^{n-p}C_{r-p}}$

Eg: In how many ways can we form a team of 11 players from 15 such that MSD & Yuvi are always included?

Soln No. of ways = $^{15-2}C_{11-2} = {}^{13}C_9$ Ans

Theorem-02 :→

No. of combination of n-distinct things taken r at a time when p particular things are always excluded = $\boxed{{}^{n-p}C_r}$

Eg: Same above ques, but this time MSD & Yuvi are always excluded.

Soln No. of ways = $^{15-2}C_{11} = {}^{13}C_{11}$ Ans

Ques A test has 6 MCQs. No. of ways in which candidate can attempt all 6 such that exactly 4 are correct is ____.

Soln _ _ _ _ _ _

${}^6C_4 \times 1 \times 1 \times 1 \times 1 \times 3 \times 3$

↑ Selecting 4 Qs need to be ~~wrong~~ correct — All Qs have only 1 correct ans. — Remaining have 3 wrong options

$= 15 \times 9 \Rightarrow 135$ Ans

Ques There are 15 players in squad out of which 7 are batsmen, 2 are keepers and 6 are bowlers. No. of ways in which 11 players team can be selected such that it has

atleast 4 bowlers
atleast 5 batsmen
atleast 1 wicketkeeper

is ____.

Soln

6 Bowlers (4)	7 batsmen (5)	2 WK (1)
4	5	$2 \Rightarrow {}^6C_4 \cdot {}^7C_5 \cdot {}^2C_2$
4	6	$1 \Rightarrow {}^6C_4 \cdot {}^7C_6 \cdot {}^2C_1$
5	5	$1 \Rightarrow {}^6C_5 \cdot {}^7C_5 \cdot {}^2C_1$

Total ways = 105 + 210 + 252 = 567 Ans

Ques Find no. of ways in which a committee can be formed from 6 Indians & 8 Foreigners which include atleast 2 Indians & double foreigners.

Soln

6 Indians	8 Foreigners
2	$4 \Rightarrow {}^6C_2 \cdot {}^8C_4$
3	$6 \Rightarrow {}^6C_3 \cdot {}^8C_6$
4	$8 \Rightarrow {}^6C_4 \cdot {}^8C_8$

Total ways = 1050 + 560 + 15 ⇒ 1625 Ans

Ques A debate has 6 girls & 4 boys, out of which a 4 member team is to be made including a captain (among those 4 members). Find no. of ways in which team can be formed such that atmost 1 boy is included in team.

Soln

6 Girls	4 Boys	
3	$1 \Rightarrow {}^6C_3 \cdot {}^4C_1 \cdot {}^4C_1$	Captain chosen
4	$0 \Rightarrow {}^6C_4 \cdot {}^4C_0 \cdot {}^4C_1$	

Total ways = 320 + 60 ⇒ 380 Ans

Number of St. Lines & Triangles :

If there are n non-collinear points in a plane then

$\boxed{\text{No. of st. lines} = {}^nC_2}$

$\boxed{\text{No. of triangles} = {}^nC_3}$

Ques There are 10 points in a plane of which 4 are collinear, and 6 are non-collinear. Find

(i) No. of straight lines

Method-1

4 Collinear	6 Non-Collinear
2	0
1	1
0	2

Total lines = ~~$^6C_2$~~ + $^4C_1 \cdot {}^6C_1 + {}^4C_0 \cdot {}^6C_2$

= 40 Ans

Method-2

No. of lines = Total − (lines that can't be formed)

⇒ No. of lines = $^{10}C_2 - {}^4C_2 + 1$

= 40 Ans

(ii) Number of Triangles

Method-1		Method-2
4 Collinear	6 Non-collinear	
1	2	No. of Δs = Total - (Δs that can't be formed)
2	1	No. of Δs = $^{10}C_3 - {}^4C_3$
0	3	$= \frac{10\times9\times8}{1\times2\times3} - 4$
		$= 116$ Ans

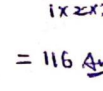
No. of Δs = $^4C_1 \cdot {}^6C_2 + {}^4C_2 \cdot {}^6C_1 + {}^6C_3$

$= 116$ Ans

Other Geometry Based articles :

1) Number of diagonals in n-sided polygon

$= \boxed{{}^nC_2 - n}$



If a polygon (n-sided) has its vertices joined together to form triangles, then

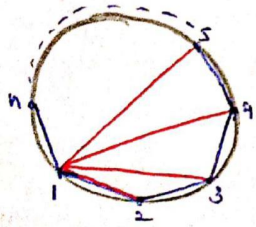
2) No. of triangles having one common side with the side of polygon = $\boxed{n(n-4)}$

3) No. of triangles having two sides common with the polygon = $\boxed{n}$

4) No. of triangles having no side common with the polygon = $\boxed{{}^nC_3 - [n + n(n-4)]}$

Ques Let $n>2$ be an integer. Suppose that there are n-Metro stations in a city located along a circular path. Each pair of stations is connected by a straight track only. Further, each pair of nearest stations is connected by blue lines, whereas all remaining pairs of stations are connected by red line. If no. of red lines is 99 times the no. of blue lines, the value of n is ____.

Sol'n Basically red lines are diagonals and blue lines are sides of n-sided polygon.



Blue lines = n

Red lines = $^nC_2 - n$

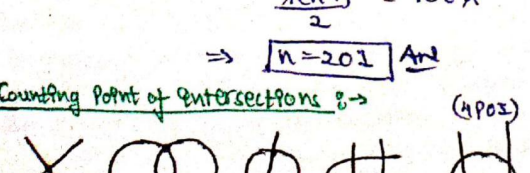
$\therefore ({}^nC_2 - n) = 99(n)$

${}^nC_2 = 100n$

$\frac{n(n-1)}{2} = 100n$

$\Rightarrow \boxed{n = 201}$ Ans

Counting Point of Intersections :→



2 Lines (1 P.O.I) | 2 Circles (2 POI) | Line-Circle (2 POI) | 2 Parabolas (4 POI) | Circle-Parabola

Permutation & Combination - 2



Line-Parabola (2 POI)

All these are max. no. of POI b/w curves.

Ques Find max. no. of POI b/w 4 lines, 3 circles and 2 Parabolas.

Sol'n Count POI b/w all possible combinations

Line-Line ⇒ $^4C_2 \times 1 = 6$

Line-Circle ⇒ $^4C_1 \cdot {}^3C_1 \times 2 = 24$

Line-Parabola ⇒ $^4C_1 \cdot {}^2C_1 \times 2 = 16$

Circle-Circle ⇒ $^3C_2 \times 2 = 6$

Circle-Parabola ⇒ $^3C_1 \cdot {}^2C_1 \times 4 = 24$

Parabola-Parabola ⇒ $^2C_2 \times 4 = 4$

Total POI = 80 Ans

Linear Permutation :

Ques 4 Boys and 4 Girls are to be seated in a line. Find number of ways

(i) If B_1 and G_1 are always together.

Bag Method ⇒

Put B_1G_1 in a bag and consider them as a single unique unit.

$\boxed{B_1G_1}\ B_2\ B_3\ B_4\ G_2\ G_3\ G_4 \Rightarrow 7$ distinct items

No. of ways = $7! \times 2!$ (Internal Arrangement of Bag)

(ii) If B_1 and G_1 are never together.

Gap Method

Arrange all items on which no cond^n is applied, leaving gap b/w each of them.

$_B_2_B_3_B_4_$

No. of ways = $6! \times {}^7C_2 \times 2!$

($^7C_2$: selection of two Gaps; $2!$: Arrangement of B_1 & G_1)

(iii) If no two girls are together

Sol'n $_B_1_B_2_B_3_B_4_$

No. of ways = $4! \times {}^5C_4 \times 4!$

(iv) If not all girls are together

Sol'n $\boxed{G_1G_2G_3G_4}\ B_1\ B_2\ B_3\ B_4$

No. of ways = Total - (All girls together)

$= 8! - (5! \times 4!)$

(v) Boys & Girls are alternate

Sol'n $B_1\ G_1\ B_2\ G_2\ B_3\ G_3\ B_4\ G_4 = 4! \times 4!$

$G_1\ B_1\ G_2\ B_2\ G_3\ B_3\ G_4\ B_4 = 4! \times 4!$

$(4!)^2 \times 2$

(vi) If there are 4 couples, in how many way can they be seated such as each couple is together.

Sol'n $\boxed{B_1G_1}\ \boxed{B_2G_2}\ \boxed{B_3G_3}\ \boxed{B_4G_4}$

No. of ways = $4! \times (2!)^4$

Ques Let n be the number of ways in which 5 Boys and 5 girls can stand in a queue such that all girls stand consecutively in the queue. Let m be the no. of ways in which 5 Boys & 5 girls can stand in a queue such that exactly 4 girls stand consecutively in the queue. Find the value of m/n.

Sol'n For n $\boxed{G_1G_2G_3G_4G_5}\ B_1B_2B_3B_4B_5$

$n = 6! \times 5!$

For m

$\boxed{G_1G_2G_3G_4}\ G_5$

$_B_1_B_2_B_3_B_4_B_5_$

$m = {}^5C_4 \times 5! \times {}^6C_2 \times 2! \times 4!$

($^5C_4$: Selection of 4 Girls to stand consecutively; $5!$: Arrange 5 Boys; $^6C_2$: selects Gaps; $2!$: Arrang bag and G_5; $4!$: Internal Arrangement in bag)

So, $\frac{m}{n} = \frac{5 \times 120 \times 6 \times 5 \times 2 \times 24}{120 \times 720 \times 2}$

$\boxed{\frac{m}{n} = 5}$ Ans

Ques Words of length 10 are formed using the letters A, B, C, D, E, F, G, H, I, J. Let x be the no. of words where no word is repeated and y be the no. of words where exactly one letter is repeated twice. Then $y/9x =$ ____.

Sol'n $x = 10!$

$y = {}^{10}C_1 \times {}^9C_8 \times \frac{10!}{2!}$

($^{10}C_1$: Selection of word to be repeated twice; $^9C_8$: Selection of remaining 8 letters for word.)

$\frac{y}{9x} = 5$ Ans

Batting Order Question :→

Ques No. of ways in which 11 players can bat, if Yuvi wants to bat before MSD and Jadeja wants to bat after MSD is ____.

Sol'n ___ Yuvi ___ ___ MSD ___ Jadeja ___ ___

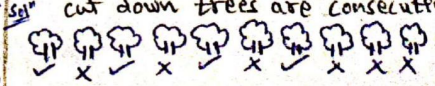
No. of ways = $^{11}C_3 \times 1 \times 8!$

($^{11}C_3$: selection of 3 positions for 3 players to bat; $8!$: Arrangement of 3 players in given orders)

Tick and Cross Method :→

Ques There are 10 trees in a row. 4 trees are to be cut. The no. of ways that no two of the cut down trees are consecutive is ____.

Sol'n



Consider,

Required items ⟶ Tick (✓)

Other items ⟶ Cross (X)

So we have, 4 Tick ✓ ✓ ✓ ✓ (All identical)

6 Cross X X X X X X (All identical)

_ X _ X _ X _ X _ X _

No. of ways = $1 \times {}^{7}C_4 \times 1$

(Arrangement of cross (can are identical) ↑ Selection of Gaps ↑ Arrangement of ticks)

Ques 4th engineer is required to visit a factory for exactly four days during the first 15 days of every month and it is mandatory that no two visit take place on consecutive days. No. of ways in which engineer can visit factory during 1st - 15th June is ____.

Sol^n Tick ⇒ 4

Cross ⇒ 11

No. of ways = $1 \times {}^{12}C_4 \times 1$

$= 495$ Ans

(Jitne item hange, usse 1 zyada gap bante hai)

#Formation of Groups/Teams :

→ This method is for n distinct items

→ Jitne amount ke groups mein divide karna hai, utne ke factorial se (n!) ko divide karo.

→ Agar 'm' no. of groups mein same amount of items hai toh (m!) se bhi divide karo.

Eg: (8 → 2, 2, 2, 1, 1) No. of ways = $\frac{8!}{(2!)^3 (2!)^2} \times \frac{1}{3!2!}$

(Because of same group strength)

Eg: (4 → 2, 2) No. of ways = $\frac{4!}{2!2!(2!)}$

NOTE :

When distinct items distributed among distinct people ⟹ Use formation of groups concept

When identical items distributed among distinct people ⟹ Use Beggar's method (Discussed ahead)

Ques The total no. of ways in which 5 ball of diff. colours can be distributed among 3 persons so that each get atleast one ball is ____.

Sol^n (5 → 1, 1, 3) + (5 → 1, 2, 2) Arrangement of groups

$\frac{5!}{(1!)^2 3! (2!)} \times 3! + \frac{5!}{1! (2!)^2 2!} \times 3!$

No. of ways $= 3! \left(\frac{120}{12} + \frac{120}{8}\right) \Rightarrow 150$ Ans

Ques Find no. of ways in which 6 person can be accomodated in 4 rooms such that each room has atleast one person.

Sol^n (6 → 2, 2, 1, 1) (6 → 3, 1, 1, 1)

$\frac{6!}{(2!)^2 (1!)^2} \times \frac{1}{2!2!} \times 4! + \frac{6!}{3! (1!)^3} \times \frac{1}{3!} \times 4!$

No. of ways $= 24 \left(\frac{720}{16} + \frac{720}{36}\right)$

$= 1560$ Ans

#Permutation of alike objects :

↳ n items mein jitne baar koi item repeat ho, utne ke factorial se (n!) ko divide karo

Eg: DADDY No. of arrangements = $\frac{5!}{3!}$

Ques Consider the word ASSASSINATION, find no. of ways of arranging the letters.

(i) No. of words using all letters

Sol^n No. of words = $\frac{13!}{3!\,4!\,2!\,2!}$ (A) (S) (I) (N)

(ii) If no vowels are together

Sol^n Vowels → A, A, I, A, I, O

_ S _ S _ S _ N _ T _ N _

First arrange all consonants, then fill vowels in the gaps.

No. of words = $\left(\frac{7!}{4!2!}\right) \times {}^{8}C_6 \times \frac{6!}{3!2!}$ Ans

(iii) Vowels are in same order

Sol^n _ _ _ _ _ _ _ _ _ _ _ _ _

A A I A I O

No. of words = ${}^{13}C_6 \times 1 \times \frac{7!}{4!2!}$ Ans

(Selection of 6 places for vowels) (Arrangement of vowels in same order)

(iv) Relative position of vowels & consonants remain same.

Sol^n Means jis jgh vowel tha uss jgh vowel hi arrange karna hai & same for consonant.

A S S A S S I N A T I O N

V C C V C C V C V C V V C

$\left(\frac{6!}{3!2!}\right)\left(\frac{7!}{4!2!}\right)$ Ans

Ques The number of 7 digit integers with sum of digits equal to 10 & formed by using digits 1, 2 and 3 only is ____.

Sol^n Case-01 ⇒ 1, 1, 1, 1, 1, 2, 3 ⇒ $\frac{7!}{5!} = 42$

Case-02 ⇒ 1, 1, 1, 1, 2, 2, 2 ⇒ $7!/4!3! = 35$

Total numbers = 42 + 35

= 77 Ans

#Permutation of objects taken some at a time :

Ques How many 5 letter words can be formed using letters of the word INDEPENDENCE.

Sol^n N→3, E→4, D→2, I→1, P→1, C→1

(I) All 5 diff ⇒ ${}^{6}C_5 \times 5!$

(II) 4 alike, 1 diff. ⇒ $1 \times {}^{5}C_1 \times 5!/4!$

(III) 3 alike, 2 diff ⇒ $\underset{(N,E)}{2} \times {}^{5}C_2 \times 5!/3!$

(IV) 3 alike, 2 alike ⇒ $\underset{(N,E)}{2} \times \underset{(D,\text{Remaining})}{2} \times 5!/3!2!$

(V) 2 alike, 3 diff ⇒ $3 \times {}^{5}C_3 \times 5!/2!$

(VI) 2 alike, 2 alike, 1 diff ⇒ ~~3 × 2 × 5~~ ${}^{3}C_2 \times {}^{4}C_1 \times \frac{5!}{2!2!}$

Total such words = Sum of above all cases.

Ques The no. of words that can be formed by 4 letters at a time from letters of word SYLLABUS such that 2 letters are diff. & 2 letters are alike is ____.

Sol^n S→2, L→2, Y→1, A→1, B→1, U→1

2 diff. 2 alike = $2 \times {}^{5}C_2 \times \frac{4!}{2!}$

$= 240$ Ans

Circular Permutation :

Circular permutations of n-distinct objects is $(n-1)!$

Linear Permutation: A B C = 3!

Circular Permutation: (ABC) = 2!

Same: A B C, A C B, B A C, B C A, C A B, C B A (Same)

(A, B, C) (A, B, C) (B, A, C) (B, C, A) (C, A, B) (C, B, A)

→ In case of Garland/necklace, the CW and ACW permutations are identical therefore

Circular Permutation in case of necklace/garland = $\frac{(n-1)!}{2}$

(A, B identical)

→ In a circular permutation, even if position of one object/person is fixed, then rest of the ques. is treated as linear permutation only.

Ques No. of ways in which 5 boys and 3 girls can be seated on a round table if a particular boy B_1 and a particular girl G_1 never sit adjacent to each other is ____.

Sol^n (G_3, B_2, B_3, B_4, B_5, G_2)

No. of ways = $5! \times {}^{6}C_2 \times 2!$

(Arrangement of 4B 2G) (Selection of Gaps)

Permutation & Combination-3

Total Number of Selections/Combinations :

$A_1 A_2 A_3 A_4 A_5$ (5 diff. apples) → No. of ways of selecting
2 apples = $^5C_2$
3 apples = $^5C_3$
⋮

$A A A A A$ (5 alike apples) → No. of ways of selecting
0 apples = 1
1 apple = 1
2 apple = 1
5 apple = 1

No. of ways of selecting	^{n} Identical things	^{n} distinct things
zero things	1	$^nC_0$
One thing	1	$^nC_1$
Two things	1	$^nC_2$
⋮	⋮	⋮
n things	1	^nC_n
Total ways of selecting =	n+1	2^n

(I) No. of ways of selecting
zero or more from n-distinct = 2^n
One or more from n-distinct = $(2^n - 1)$

(II) No. of ways of selecting
zero or more from n-identical = (n+1)
One or more from n-Identical = n

Ques A basket has 4 apples, 3 Mangoes, 2 bananas, 2 oranges.

(i) Consider fruits of same specie to be alike and find the no. of ways that

(a) atleast one fruit is selected

Sol^n A A A A (4) M M M (3) O O (2) B B (2)
(1)(1)(1)(1) (1)(1)(1)(1) (1)(1)(1) (1)(1)(1)
No. of ways = Total − 1 (When none is selected)
= (4+1)(3+1)(2+1)(2+1) − 1
= 119 Ans

(b) atleast one fruit of each kind is selected

Sol^n Don't count 'zero wala' case for all fruits.
No. of ways = (4)(3)(2)(2) = 48 Ans

(ii) Consider fruits of same specie are also diff. then find the no. of ways that

(a) atleast one fruit is selected

Sol^n No. of ways = Total − 1
= $(2^4)(2^3)(2^2)(2^2) - 1$
= 2047 Ans

(b) atleast one fruit of each kind is selected

Sol^n No. of ways = $(2^4-1)(2^3-1)(2^2-1)(2^2-1)$
= (15)(7)(3)(3) = 945 Ans

Number Theory :

(I) Total Number of Divisors :→

→ Simply prime factorise the given number and apply concepts of selection of Identical objects (considering all powers of a prime no. as that amount of identical object).

Eg : N = 12

2	12
2	6
3	3
	1

$N = 2^2 . 3^1$
No. of divisors = (2+1)(1+1) = 6
i.e. Divisors ⇒ 1, 2, 3, 4, 6, 12

(II) Number of even divisor :→

→ For divisor to be even, 2 must be present
→ Don't consider the case where zero '2' is selected

Eg N = 12
No. of even divisors = (2)(1+1) = 4
(2, 4, 6, 12) ←

(III) Number of odd divisors : → Total − (Even divisors)

(IV) Sum of divisors :→

→ Prime ki successive powers ka add karo, for each particular prime no.
→ Then simply multiply these additions.

Eg N = 12
Sum of divisors = $(2^0+2^1+2^2)(3^0+3^1)$
= (7)(4) = 28 Ans

(V) Number of ways of representing N as product of 2 Numbers :→

= (No. of divisors + 1)/2, if N → Perfect sq.

= (No. of divisors)/2, if N → not a perfect sq.

Ques For $N = 2^5 . 3^4 . 5^1 . 7^2$, find
(i) Total no. of divisors = (6)(5)(2)(3) = 720
(ii) No. of Proper divisors = 720 − 2 = 718
(All divisors except 1 & No. itself)
(iii) No. of even divisors = (5)(5)(2)(3) = 600
(iv) No. of odd divisors = 720 − 600 = 120
(v) No. of divisors, divisible by 5 = (6)(5)(1)(3) = 630
(Not considered case when zero '5' were selected)
(vi) No. of divisors divisible by 10 = (5)(5)(1)(3) = 525
(÷2 & ÷5)
(vii) No. of divisors divisible by 2 but not by 4.
Sol^n ~~Selection~~ Select only the case of (2^1)
No. of divisors = (1)(5)(2)(3) = 120 Ans

(viii) Sum of all divisors
$= (2^0+2^1 \ldots 2^5)(3^0+3^1 \ldots 3^4)(5^0+5^1 \ldots 5^1)(7^0+7^1+7^2)$

(ix) Sum of even divisors
Min. one power of 2 must be present
$= (2^1+2^2 \ldots 2^5)(3^0+3^1 \ldots 3^4)(5^0+5^1 \ldots +5^1)(7^0+7^1+7^2)$

(x) Sum of odd divisors
$= (2^0)(3^0+3^1 \ldots 3^4)(5^0+5^1 \ldots +5^1)(7^0+7^1+7^2)$

Ques $N = 2^x 3^y 5^z$ is a natural no. such that $y+z = 5$ and $y^{-1}+z^{-1} = 5/6$, $y > z$, then no. of odd divisors of N is ____.

Sol^n $y+z=5$ ⋮ $\frac{1}{y}+\frac{1}{z} = \frac{z+y}{yz} = \frac{5}{6} \Rightarrow yz = 6$
These eq^ns simply (alongwithy > z) that y = 3 & z = 2
No. of odd divisors = (1)(3+1)(2+1) = 12 Ans

Summation of Numbers :

Shortcut with no condition :→

(Jitne digit ke no. ka sum karna hai, utne '1') × (Sum of digits given to form numbers) × (Jitne digit no. hai, usse ek kam ka factorial)

Eg : Find sum of all 3 digit numbers formed by 1,2,3 (w/o repetition)

Sol^n Sum = 111 × (1+2+3) × (3−1)!
= 111 × 12 ⇒
12
12
1 12
1332 Ans

Ques Find sum of numbers greater than 10000 formed by digits 1,3,5,7,9 w/o being repeated.

Sol^n Sum = 11111 × 25 × 4!
= 11111 × 600 ⇒ 6666600 Ans

Ques Sum of all 3 digit numbers that can be formed using 1,2,3,4 (w/o repetition) is ____.

Sol^n Here, we will have to form cases & then add.

(1,2,3) ⇒ 111 × 6 × 2!
(1,2,4) ⇒ 111 × 7 × 2!
(1,3,4) ⇒ 111 × 8 × 2!
(2,3,4) ⇒ 111 × 9 × 2!
⇒ 111 × 2! (6+7+8+9)
= 222 × 30
= 6660 Ans

Shortcut when zero is included :→

↳ Simply apply shortcut for given 'digit number' and then minus the sum of numbers formed excluding zero.

Ques Find sum of 3 digit no. formed by 0,1,2.

Sol^n (0,1,2) − (1,2)
[111 × 3 × 2!] − [11 × 3 × 1!] = 633 Ans

Shortcut when digits are repeated :→

↳ Last wale factorial ko repetition wale factorials se divide karo.

Ques Find sum of 4 digit numbers that can be formed using 1,1,3,4.

Sol^n Sum = 1111 × 9 × $\frac{3!}{2!}$ (1 repeated two times)

= 1111 × 27

Sum ⇒ 29997 **Ans**

Ques The sum of all 4-digit numbers that can be formed with digits 1,2,2,3 is ____.

Sol^n Sum = 1111 × 8 × $\frac{3!}{2!}$ ⇒ 1111 × 24

Sum ⇒ 26664 **Ans**

Distribution of alike objects/Beggar's Method :

Eg : 6 identical coins need to be distributed among 3 beggars. Find no. of ways to do this.

Sol^n Jitne beggars hai, usse ek kam sticks assume karlo which will help to distribute coins, and then simply distribute (coin+stick)

OOOOOO | | = $\frac{8!}{6!\,2!}$ = 28

6 coins (identical) 2 sticks (identical)

Ques Find no. of ways of distributing 7 identical tablets among A, B and C such that each recieve atleast one tablet.

Sol^n Pehle, jitne tabs pe condition hai, utne distribute karo.

A B C ; No. of ways = $\frac{(4+2)!}{4!\,2!}$

1 1 1 ;

= 15 **Ans**

Use of Beggar's method in Sum/Product questions :→

Ques Find no. of natural solutions of equation $x+y+z=102$, where $x,y,z \in N$

Sol^n Since $x,y,z \in N$, each will have min. value of (1)

No. of solutions = $\frac{(99+2)!}{99!\,2!} = \frac{101 \times 50}{}$

= ~~4850 Ans~~ 5050 **Ans**

Ques Number of non-negative integral solution of inequality $x+y+z+t \leq 30$ is ____.

Sol^n whenever, sum is less than/less than equal to, assume a false beggar and solve.

Ex $x+y+z+t+(\omega) = 30$

30 coins 4 sticks

No. of solutions = $\frac{34!}{30!\,4!}$ = 43676 **Ans**

Ques Find no. of integral sol^n of $x+y+z+t=16$ such that $x \geq 2$, $y \geq 2$, $z \leq 3$, $t \geq -2$

Sol^n $x+y+z+t = 16$ (with conditions)

$(x-2) \geq 0 \mid (y-2) \geq 0 \mid (z+3) \geq 0 \mid (t+2) \geq 0$ (convert $\leq \to \geq$)

$(x-2)+(y-2)+(z+3)+(t+2) = 16$ (No cond^n)

$x+y+z+t = 15$ → 15 coins, → 3 sticks

No. of solutions = $\frac{18!}{15!\,3!}$ = 816 **Ans**

Ques Find solutions of $x+y+z+4\omega = 20$, such that $x,y,z,\omega \in$ +ve integers.

Sol^n When,

$\omega = 1 \Rightarrow x+y+z = 16 \Rightarrow \frac{18!}{16!\,2!} = 153$

$\omega = 2 \Rightarrow x+y+z = 12 \Rightarrow \frac{14!}{12!\,2!} = 91$

$\omega = 3 \Rightarrow x+y+z = 8 \Rightarrow \frac{10!}{8!\,2!} = 45$

$\omega = 4 \Rightarrow x+y+z = 4 \Rightarrow \frac{6!}{4!\,2!} = 15$

Total no. of sol^{n}s = 304 **Ans**

Ques $(x_1+x_2+x_3)\cdot(y_1+y_2) = 77$, find +ve integral solutions.

Sol^n 77

1 × 77]

77 × 1] ⇒ Not possible because sum of two +ve integers can never be 1.

7 × 11

11 × 7

(i) $x_1+x_2+x_3 = 11$ and $y_1+y_2 = 7$

$\frac{10!}{10!\,2!} = 78$ $\frac{8!}{7!\,1!} = 8$

(ii) $x_1+x_2+x_3 = 7$ and $y_1+y_2 = 11$

$\frac{9!}{7!\,2!} = 36$ $\frac{12!}{11!\,1!} = 12$

No. of solutions = (78 × 8) + (36 × 12) = 1008 **Ans**

(Sabka atleast 1 dena padega as +ve integral sol^n)

(i) $x_1+x_2+x_3 = (11-3)$ and $y_1+y_2 = (8-2)$

8 coins 2 sticks 6 coins 1 stick

$\frac{10!}{8!\,2!} = 45$ $\frac{6!}{5!\,1!} = 6$

(ii) $x_1+x_2+x_3 = (7-3)$ and $y_1+y_2 = (11-2)$

4 coins 2 sticks 9 coins 1 stick

$\frac{6!}{4!\,2!} = 15$ $\frac{10!}{9!\,1!} = 10$

Total sol^{n}s = (45 × 6) + (15 × 10) = 420 **Ans**

Ques Total +ve integral sol^n of $xyz = 24$ is ____.

Sol^n Prime factorise the number and distribute each prime no. acc. to Beggar's method.

$N = 24 = 2^3 \cdot 3^1$

3 coins 2 sticks 1 coin 2 sticks

$\frac{5!}{3!\,2!} \times \frac{3!}{1!\,2!}$

No. of sol^{n}s = (10)(3) = 30 **Ans**

Dearrangement :

If n things are arranged in a row, the no. of ways they can be dearranged so that 'r things occupy wrong places' while '(n−r) things occupy their original places' is

= $^nC_r \cdot D_r$

where, $D_r = r!\left(1 - \frac{1}{1!} + \frac{1}{2!} - \frac{1}{3!} \cdots (-1)^r \frac{1}{r!}\right)$

odd ke aage → −ve

even ke aage → +ve

Remember ⇒

$D_0 = 1$ $D_2 = 1$ $D_4 = 9$

$D_1 = 0$ $D_3 = 2$ $D_5 = 44$

Ques A person write letters to 5 friends. Find ways in which letters are posted such that

(i) All letters are in wrong envelopes.

Sol^n No. of ways = $^5C_5 \cdot D_5$

= 1 × 44 = 44 **Ans**

(ii) Atleast 3 of them are in wrong envelope

Sol^n No. of ways = $^5C_3 D_3 + {}^5C_4 \cdot D_4 + {}^5C_5 \cdot D_5$

= (10)(2) + 5(9) + 1(44)

= ~~Ans~~ 109 **Ans**

Ques 6 cards and 6 envelopes are numbered 1 to 6, and cards are to be placed in envelope so that each ~~card~~ envelope contains exactly one card and no card is placed in the envelope bearing the same no. and moreover the card numbered 1 is always placed in envelope numbered 2; then no. of ways it can be done is ____. [JEE Adv 2014]

Sol^n Case-01 : Card 2 is placed in envelope 1 as well

No. of ways = $D_4 = 9$

Case-02 : Card 2 is not placed in envelope 1

No. of ways = $D_5 = 44$

Total no. of ways = 53 **Ans**